

Ravi Prakash Srivastava

+91-9431334691 raviprakashshrivastav7@gmail.com er-ravi-srivastava.github.io

Research Interests

Large Language Models (LLMs), AI Agents, Trustworthy AI, NLP, Adversarial Machine Learning, Computer Vision.

Education

Indian Institute of Information Technology Ranchi (IIIT Ranchi), India M.Tech (Data Science & AI) <i>Institute of National Importance, Govt. of India</i>	August 2024 – May 2026 CGPA: 8.79/10
North Eastern Hill University (NEHU), Shillong, India B.Tech (Information Technology) <i>Central University, Govt. of India</i>	August 2018 – June 2022 CGPA: 7.85/10

Experience

Software Engineer SS Software Solutions LLC, Hyderabad, India March 2026 – Present

- Defined and enforced ML pipeline standards encompassing model versioning (MLflow/DVC), structured logging (JSON schema-validated with trace IDs for distributed tracing), and observability instrumentation (metrics, alerts, drift detection), reducing incident diagnosis time by 40% and improving pipeline reliability across 3+ production models.
- Standardized ML pipeline governance across versioning, logging, and monitoring layers, cutting untracked model deployments to zero and reducing mean-time-to-debug (MTTD) by ~35%; applied STRIDE-based Threat Modeling to identify 8+ design-level risks including data poisoning vectors and model inversion attack surfaces.
- Automated security testing by integrating SAST tools (Bandit, Semgrep) and dependency vulnerability scanning into CI/CD pipelines (GitHub Actions), reducing manual security review effort by ~60% and accelerating release cycles.

Machine Learning Intern C-DAC (Govt. of India), Hyderabad, India August 2022 – February 2023

- Performed large-scale data preprocessing and feature engineering on student activity logs (attendance, assignments, quizzes) to support predictive modeling.
- Implemented and benchmarked multiple machine learning models, including Decision Trees, Random Forests, and Support Vector Machines, for academic performance prediction.
- Evaluated model performance using cross-validation and metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC, identifying the best-performing model for early prediction of at-risk students.

Research Projects

- Adversarial Prompt Injection Attacks on LLM: Cryptographic Key Leakage and Defense Strategies**
Under Publication at International Conference on Advanced Computing and Applications (ICACA-2026).
- IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions using Deep Learning**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2025.2521688>
- Deepfake Video Detection Using Face-Centric Processing and Frame Sampling**
Under Publication International Conference on Advanced Computing and Applications (ICACA-2026).
- Balanced Few-Shot Episodic Learning for Accurate Retinal Disease Diagnosis**
(arXiv: <https://arxiv.org/abs/2512.04967v1>)
- Smart City Vehicle Accident Monitoring and Detection System Using MEMS, GSM, GPS Raspberry Pi 4**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2022.2043787>

Projects

AI-Powered Vulnerability Management Platform (SAST + DAST) Live Demo

- Automated SAST scanning using Semgrep for static code analysis and early vulnerability detection.

- DAST testing covering SQLi, XSS, CSRF, and common web vulnerabilities in live environments.
- Interactive Streamlit-based dashboard for real-time vulnerability visualisation and reporting.
- Vulnerability prioritization engine with security remediation recommendations.
- CI/CD and DevSecOps-friendly architecture supporting seamless pipeline integration.
- Dockerized deployment for scalability, portability, and consistent environment management.

Technical Skills

- **Technologies:** Python, JavaScript.
- **Databases:** SQL, MariaDB, MySQL, relational schema design.
- **Machine Learning & AI:** Deep Learning, Machine Learning, Computer Vision, Image Processing, Deepfake Detection, Large Language Models (LLMs), Hugging Face Transformers, PyTorch, Scikit-learn, OpenCV.
- **MLOps & DevOps:** Git, Docker, GitHub Actions (CI/CD), MLflow, DVC.

References

- **Dr. Ajay Kumar**
Chairman membership, Institution of Electronics and Telecommunication Engineers, India
Email: chairman.membership@iete.org
- **Prof. Md. Iftekhar Hussain**
Professor of Information Technology, North-Eastern Hill University, India
Email: ihussain@nehu.ac.in